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Smart Traffic Light Control System with Emergency Vehicle Detection and Boom Barriers Using Arduino

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***Abstract* — Urbanization and modernization have significantly increased traffic congestion, creating challenges for daily commuters and emergency services. The traditional traffic light system, which runs on a fixed timer, results in commuters often being stuck in unnecessary traffic, wasting time and fuel, while emergency services face delays costing crucial time in emergency situations. Additionally, intersection-related accidents account for 40% of all traffic incidents and nearly half of all serious injuries, with red-light running causing approximately 165,000 accidents annually, resulting in 700 to 800 fatalities. This study investigates the use of an Arduino-driven traffic control system to integrate real-time traffic density monitoring, emergency vehicles prioritization, and enforcement mechanisms such as boom barriers to enhance intersection safety and optimize traffic flow. The system demonstrated the ability to detect the density of vehicles in each lane of the road using ultrasonic sensors and prioritize emergency vehicles using sound sensors. These sensors provide data that serve as input to the Arduino which results in boom barriers to function accordingly. The prototype simulated traffic challenges at traffic intersections with approaching emergency vehicles. Results showed that a scaled prototype demonstrated reduced congestion, improved intersection safety, and enhanced emergency vehicle response times. This prototype system offers a cost-effective, scalable solution to modern urban traffic management challenges. Future work will focus on developing the prototype on a larger scale, addressing system maintenance challenges, and exploring its usage in diverse urban environments to further enhance safety and efficiency.**

Keywords— ***Smart traffic light; Arduino-based traffic control; Boom Barriers; Emergency vehicle prioritization; Traffic density monitoring***

I. INTRODUCTION

In today's world of modernization and urbanization, traffic congestion has become a pressing issue in every major city across the globe. About 68% of the world's population will be living in urban areas by 2050, according to a projection of world urbanization, undertaken by the United Nations (UN) in 2018 [1]. The increasing population along with its economic activities in urban areas contribute to the higher vehicle usage, leading to more congestion [2]. Current statistics reveals that an average person spends around 4-6 months of his/her life by simply waiting for a green light during traffic [3]. In New York City, the wait time due to traffic congestion is costing an average person 101 hours per year [4]. The traditional traffic light system, which runs on a fixed timer without considering the real-time traffic, causes more congestion, higher fuel use and more pollution. Vehicles often sit on red lights even when the junction is clear (no car passing) which causes start-stop driving patterns which reduces the lifetime of vehicles.

Moreover, traffic congestion also causes delays in emergency services such as fire, ambulance, and police. Intersection-related accidents have skyrocketed over the past five years i.e. 40% of all traffic incidents and almost half of all serious injuries [3]. Red-light running alone is responsible for around 165,000 accidents every year, tragically resulting in 700 to 800 fatalities [5]. Numerous reports have been conducted to address this issue, aiming to reduce traffic congestion and minimize accidents as well as improve the response time for emergency vehicles.

Section 2 provides a review of the relevant literature. Section 3 outlines the research question and methodology. Section 4 presents the results. Finally, Section 5 concludes the paper.

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II. LITERATURE REVIEW

Effective traffic management systems have been the focus of extensive research, with various studies proposing innovative solutions to optimize traffic flow and reduce congestion at intersections. These studies provide valuable insights into optimizing traffic flow and reducing delays at intersections. However, there are still areas, such as prioritizing emergency vehicles and enhancing intersection safety, that can be further explored. This section reviews relevant studies, highlighting their contributions, and identifying opportunities to build on their work.

Several studies have explored the use of sensor-based systems and Arduino technology to optimize traffic light operations at intersections. For example, Qasim et al. [6] used traffic light data collected from sensors, programmed to Arduino, that are installed at intersections. It involves LDR (Light Dependent Sensor) and Laser sensor where Laser sensor is fitted to shine onto LDR. When a vehicle passes through the radar of Laser Sensor, Laser Beam light intensity changes and the data is processed by microcontroller (i.e. Arduino) which then directs the optimization of the timing of lights. While this approach offers several advantages it does not fully address the issue of reducing the wait time of emergency vehicles. Further study may be necessary to ensure its effectiveness in severe congestion.

Similarly, Abubakar et al. [7] sought to investigate congestion by assessing density-based traffic lights which are capable of changing the timings of signals on the basis of the traffic collected in real time. The basic features of this system are sensors that detect the vehicle density, IR (Infrared) sensors and a microcontroller based on Arduino Uno to adjust the signals relating to traffic. They conclude that this system can dramatically decrease delays for drivers at intersections and enhance the overall traffic control efficiency. However, it does not address the faster response for emergency vehicles or safer intersections. These areas can be further explored to build upon the foundations laid by these studies.

Rai et al. [8] propose the idea to solve the traffic congestion by allocating the lights timing in relation to the density of the road. Here an IR sensor which consists of an IR transmitter, light emitter and reflector is placed on all sides of the junction. When an IR sensor detects the number of vehicles passing through the road, it conveys the data as an input to Arduino. Then the Arduino counts the total number of vehicles on all sides of the junction and turns the green light to the side which has high vehicle density. Although this model proves somewhat effective to reduce traffic congestion, it faces significant challenges related to sensor sensitivity and has very limited applications.

Similarly, Prathamesh, et al. [9], propose a concept to solve the traffic by changing the signals based on density of the lanes. The researchers used the piezoelectric sensor, along with an operation amplifier. Here the Piezoelectric sensor is placed underground to operate as a counter for every vehicle passing through. When a vehicle is passing above the piezoelectric sensor which is based on piezoelectric principle, the electric charge will be generated in response to mechanical energy. The produced electric charge from the sensor is transmitted as an input to the Microcontroller (Arduino). Then Arduino will command the lights (LEDs) to function according to the density of the lane detected by the piezoelectric sensor. Although the study provides an effective solution to control the timings of light based on density, it does not touch the issue of emergency vehicles prioritization.

G. Iswarya, et al. [10] put forward the idea of prioritizing the emergency vehicles during the congestion to save crucial time during an emergency. The researchers used two sound sensors to manage traffic for emergency vehicles. It implemented two Arduinos and two sound sensors for each lane. Sound Sensor (SS) I is connected to Arduino Uno I, and Sound Sensor II is connected to Arduino Uno II. SS I is placed 100 meters from the traffic light, while SS II is placed closer to the traffic light, with communication established through the XBEE protocol. When SS I detects an emergency vehicle, the driving lane turns green for 120 seconds while the remaining lanes turn red. Once SS II detects the vehicle, the traffic light is delayed by two seconds to allow the vehicle to pass through completely. Although this study effectively addresses the challenge of emergency vehicle prioritization in heavy traffic, it is limited to emergency vehicles with sirens and does not account for other vehicles. Further research and data collection are required to expand its applicability.

Building on this focus on emergency vehicles, Low & Bashi [11], on the “Smart Traffic Light Monitoring System for Emergency Using Arduino,” uses an Arduino microcontroller to tackle the issue on Traffic congestion. The researchers concluded from their previous data that RFID sensors are often the best technology for regulating traffic signals for emergency vehicles. Their study combined the data from other research and developed a prototype named the Smart Traffic Monitoring System using Arduino Uno. Additionally, the study gathered data through a survey using Google Forms. The respondents, aged between 16 and 46, expressed that they often panic upon hearing an emergency siren. The students chosen for the survey had diverse driving experiences, and the survey was relatively neutral compared to existing traffic systems. However, while this research identified the use of sensors, it didn’t fully address the problem of traffic congestion. It did help solve the issue of emergency vehicle delays but fell short of tackling the broader problem of congestion. To better address this issue, traffic lights should be controlled according to vehicle density while still prioritizing emergency vehicles. Further investigation is required to address this gap, particularly for the 16–29 age group.

Expanding on density-based systems, Saranya et al. [12] introduced a traffic control system using an Arduino Mega platform integrated with IR sensors and LEDs (Red and Green). The system detects vehicle density in each lane using IR sensors and dynamically adjusts traffic signals based on programmed logic. It prioritizes heavily congested lanes while alternating green lights after two cycles to ensure balanced traffic management. The design focuses on minimizing delays at intersections by adapting to real-time traffic conditions. However, the system has notable limitations. It excludes yellow

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lights, which are crucial for smooth signal transitions, and relies on IR sensors with a limited range, potentially leading to inaccuracies over longer distances. Additionally, the study focuses solely on traffic density and lacks provisions for handling emergency vehicles arriving from lower-density lanes, limiting its practical application in critical scenarios.

Zaldívar-Colado et al. [13] proposed a system that aims to optimize emergency response times by allowing emergency vehicle co-drivers to remotely control traffic lights using an Android-based mobile application. The working method consists of equipping each traffic light with an Arduino microcontroller and an Ethernet Shield, thereby enabling it to communicate with an Android mobile application. The GPS-enabled app detects the vehicle's location when it is going to cross an intersection, allowing the co-driver to say which way the traffic lights should go "Go Straight" or "Turn Left or Right" The app sends commands through the Internet to the Arduino over the Ethernet Shield. The Arduino interprets the commands and blue traffic lights for the fire engine, and red for other ways, preventing an accident. The lights then resume their normal sequence after the emergency vehicle passed. This method is no doubt very successful when it comes to curb delays and giving emergency vehicle priority, but it does not offer real-time traffic density monitoring and strong security measures.

Lastly, Dhole et al. [14] proposed a Smart Traffic Signal (STS) system using an Arduino platform and ultrasonic sensors to address traffic signal violations at intersections. The system detects vehicles crossing the stop line during a red signal using ultrasonic sensors, triggering a camera to capture the vehicle's image, which is stored with a timestamp for record-keeping. The design is practical for dense traffic areas, cost-effective, and requires no modifications to vehicles, making it easy to implement. However, the system only focuses on detecting violations without physically preventing vehicles from entering the intersection, causing the likelihood of accidents. A preventive mechanism, such as boom barriers, is recommended to enhance intersection safety by physically stopping vehicles.

In conclusion, the review studies highlight significant advancements in Arduino-driven traffic management systems, including density-based adjustments and emergency vehicles prioritization. Building on these insights, this study integrates real-time traffic density monitoring, emergency vehicle prioritization, and enforce safety mechanisms such as boom barriers to enhance intersection safety and optimize traffic flow.

III. METHODOLOGY

This study seeks to answer the research *"How can a sensor-based, Arduino-driven traffic control system, integrated with boom barriers, improve traffic flow and emergency response times?*" This research explores the potential of integrating traffic density monitoring, emergency vehicle prioritization, and enforcement mechanisms such as boom barriers to mitigate congestion and enhance intersection safety. The methodology of this study involves system integration, the designing the prototype and testing through various scenarios of a smart traffic control system.

The proposed system uses dual Arduino microcontrollers to process real-time data from ultrasonic sensors for traffic density monitoring and sound sensors for emergency vehicles detection. Additionally, a Bluetooth manual control allows manual human intervention when necessary. Boom barriers are also integrated in order to stop vehicles during red signals to reduce the intersection related accidents. Communication between Arduino is facilitated using the XBEE protocol, ensuring reliable and efficient data transfer**.** Fig.1: shows the logical flow of Smart Traffic Control System. The system continuously monitors for emergency vehicle detection. Once an emergency vehicle is detected, the system prioritizes actions (i.e. extending the green light and raising boom barriers) for the emergency vehicle, bypassing regular traffic density monitoring until the emergency has been handled.

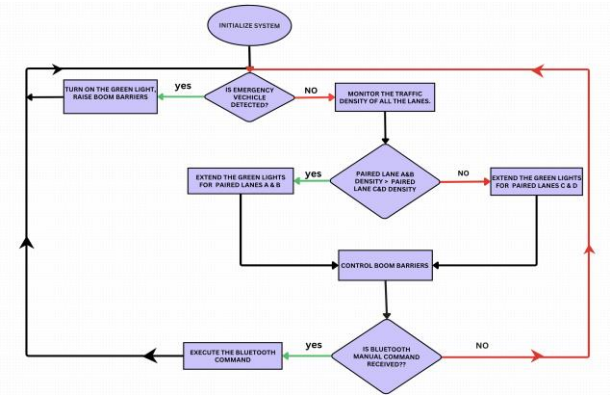


Figure 1: Logical flow of Smart Traffic Control System

The prototype was designed by integrating the hardware component and programming it through the Arduino IDE programming tool. Here to Start, the lanes are organized into two groups. Horizontal lane as lane A and B and Vertical Lane as Lane C and lane D. The vertical lane and horizontal lane are programmed to alternate with each other.

Sound sensors are placed 11 cm away from intersection on each lane to detect the siren of emergency vehicles and ultrasonic sensors are placed 3 cm away from intersection on each lane to monitor the traffic density. In addition, the Servo motor is installed in each lane to open barriers during green light and close barriers during red light.

All these sensors and traffic lights are connected to two microcontrollers named as Arduino and are sync with their respective lanes. Arduino is regarded as the brain state of the whole system which will receive data continuously from the sensor as the form input and process it as programmed. For cost-effective prototype the all the components on lane A and C are connected to Arduino I and all the components of Lane B and D are connected to Arduino II. The communication between both Arduino is established through XBEE protocol which ensures synchronization of lane A and lane B, along with Lane C and D.

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Through Arduino IDE the prototype is programmed to run on a Normal Traffic flow i.e. 6 seconds of red and green lights and 3 seconds of yellow synched with paired lanes alternating to opposite lanes while the boom barriers are also in sync to their respective lanes. In case of emergency vehicles, the sound sensor detects the siren of vehicles and then extends the green light of the lane that the sound sensor has detected. Also, it extends the red light of opposite lanes by 6 seconds. Then the Ultrasonic sensor checks the combined density of paired lanes and compares them with opposite lanes to determine the density prioritization. The paired lane having higher density gets priority by green light extending for 6 seconds.

Also, Bluetooth manual control has been added by connecting the pins of the Bluetooth module with Arduino I. This control has been added such that it can control the prototype override the operation manually in the case of breakdown or any external issues.

The prototype was then tested extensively under various scenarios. Through extensive testing we calculated the sensor response time, Synchronization timings and Communication of two Arduino boards. Furthermore, the testing was carried out for an extensive period to calculate the system durability and reliability.

Ensuring all the connection is secure and done properly through both Arduino design prototypes are created to save life in emergencies, save the time in emergency situations and reduce intersection accidents significantly. The design of "Smart Traffic Light Control System with Emergency Vehicle Detection and Boom Barriers Using Arduino" is meant to address all existing issues of congestion while prioritizing the emergency vehicles and significantly reduce accidents through the addition of Boom barriers. Through prototype design and circuit design this research hopes to contribute to address one of the burning problems of today's world.

IV. RESULTS

The designed prototype effectively demonstrated the functionality and efficiency of smart traffic lights under the various test scenarios. The prototype successfully prioritized emergency vehicles, managed the traffic density and ensured synchronized communication between all components. Each test case focused on a specific functionality and scenario as detailed below.

A. Test case I: Normal Traffic Flow Synchronization

In the first scenario, all the traffic lights on Lane A, Lane B, Lane C and Lane D are operated in a synchronized manner, Vertical Lanes (A and B) and Horizontal Lanes (C and D) are synchronized together. Each direction follows a normal traffic flow i.e. 6 seconds each for red and green lights and 3 seconds for yellow light and is opposite to the vertical lane A and B which runs on the same timing of normal flow. The prototype demonstrated effective synchronization and timing across all lanes. Fig 2: shows implementation of synchronized traffic lanes.

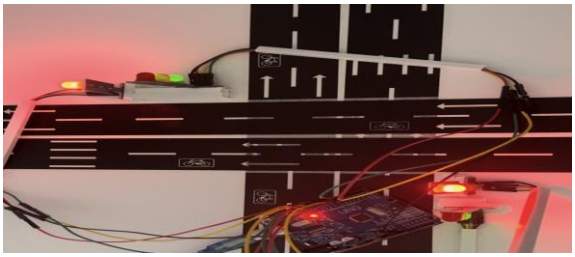


Figure 2: Synchronization of paired Lanes

B. Test case II: Emergency Vehicle Detection

A sound sensor placed on lane B, 11 cm away from the intersection in the designed prototype. An emergency siren was played with the speaker connected to the vehicle. Upon detecting the siren, the sound sensor detects the siren from the vehicle and extends the green lights of Lane A and Lane B by 6 seconds ensuring complete passage of the emergency vehicle. The test validated the system's ability to detect emergency vehicles and adjust signal timing to facilitate their passage effectively.

C. Test Case III: Emergency Vehicle Overriding Density-Based Prioritization

In this scenario, ultrasonic sensors are placed 4 cm away from each intersection. Simultaneously, the sound sensor on Lane B detected the Emergency, while the ultrasonic sensors on Lanes C and D combined identified higher traffic density than Lanes A and B. Observation was made that the designed prototype correctly prioritized an emergency vehicle overriding the density-based prioritization logic. This test demonstrated the system's capability to prioritize emergency vehicles effectively while maintaining efficient traffic flow management.

D. Test IV: Boom Barriers Synchronization

This test evaluated the synchronization of boom barriers with traffic signals. Boom barriers were placed 1.8 cm away from the intersection. When the lights turn red on Lane A and Lane B, the boom barriers lowered down on both lanes to stop red light crossings. Simultaneously, the barriers on Lane C and Lane D raised up during the green light cycle. This ensured orderly traffic movement and improved intersection safety by physically enforcing signal compliance.

E. Test Case V: Manual Control Via Bluetooth

In this test, a smartphone was connected to the prototype's Bluetooth module. A manual command was sent to extend the green light by 6 seconds on Lane A and to open the associated

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boom barriers. The prototype responded to the manual command within 1.2 seconds. Furthermore, after disengaging the manual control the prototype resumed its normal operation within 1.2 seconds. This test demonstrated the prototype's responsiveness and adaptability to manual intervention.

The results from the test cases confirm the effectiveness of the designed prototype in managing traffic flow, prioritizing emergency vehicles, and ensuring intersection safety through boom barrier synchronization. The system exhibited high responsiveness and adaptability, making it a promising solution for modern traffic management challenges.

V. Conclusion

The "Smart Traffic Light Control System with Emergency Vehicle Detection and Boom Barriers Using Arduino" Prototype successfully demonstrated its ability to address key challenges in modern traffic management. We have demonstrated through various test cases that the prototype system effectively prioritizes emergency vehicles, adjusts the traffic light based on real-time density and ensures synchronized operation across pair lanes. Additionally, the integration of boom barriers has enhanced the intersection safety by preventing red-light crossings.

The designed prototype relies on a sound sensor heavily which sometimes can be affected by environmental noise which might lead to false detection and incorrect output. Also, in the high-speed intersection the sensor might not provide accurate readings to process. These are some of the limitations of the designed prototype.

Expanding the research and prototype can be done by incorporating renewable energy like solar panels which will make the designed prototype more sustainable. Also, developing the centralized app for remote control and overriding the operations through the app will add flexibility to the system. Also, Further work will include addressing real-word implementation challenges, maintenance and costs evaluations.

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